

Introduction to Computational Stochastic PDEs, CUP, 2014

Here are the typos/errors that we know about in the first edition. Detailed corrections to MATLAB codes are given on-line ¹. Let us know if you find anymore.
8th August 2024

Chapter 1. **p17** In Definition 1.60, the Hilbert–Schmidt norm should be defined as

$$\|L\|_{\text{HS}(U,H)} := \left(\sum_{j=1}^{\infty} \|L\phi_j\|^2 \right)^{1/2},$$

(the norm on the right is the one in H not in U).

p21 Lemma 1.78 (Dini’s lemma) requires additionally the assumption that f is continuous (to guarantee that $f(x) - f_n(x) \geq \epsilon$ for the limit point x). In the application of Dini’s lemma for the proof of Theorem 1.80 (Mercer’s theorem), this holds true for $g(x) = G(x, x)$.

Chapter 2. **p42** Figure 2.1 is misprinted.

p56 In the first displayed equation of the proof, the first term in the integrand should be $p(x)(e(x)')^2$ (the $e(x)$ has one too many dashes).

p79 The right-hand side of (2.109) should be $c_1|\hat{u} - I_h\hat{u}|_{H^2(\Delta^*)}^2$.

Assumption 2.64 in general can only be verified for constant boundary data g ; H^2 -regularity results usually include an extra term on the right-hand side to account for $g \neq 0$ – see (Renardy and Rogers, 2004, Theorem 8.53) or (McLean, 2000, Theorem 4.10).

Chapter 3. **p116** `meshgrid` is misused in Example 3.40 and Algorithm 3.6. See `exa_3.40.m`.

p132 Following the comment on p487 below, the last line of the proof should be

$$\|u(t_n) - \tilde{u}_n\| \leq E \exp(Ln\Delta t).$$

Chapter 4. **p159** Nensen’s inequality $\rightarrow$ Jensen’s inequality (in proof of T4.58 (iii)).

p179 Bayes’ theorem is due to the Reverend Thomas Bayes and the apostrophe is written after and not, as in Exercise 4.11, before the s . In the same exercise, $p_{X,Y}$ is incorrectly defined and it should be

$$\mathbb{P}(X = x_k, Y = y_j) = P_{X,Y}(k, j)$$

(interchange j and k).

Chapter 5. **p185** In the second displayed equation on the left-hand side, delete the comma

p203 By "If the process is Gaussian" in Theorem 5.29, we mean X is Gaussian in the sense of Definition 4.38, which immediately implies that the ξ_j are Gaussian. It can be shown that, if all the finite-dimensional distributions are Gaussian, and sample paths $X \in L^2(\mathcal{T})$, then X is also Gaussian in the sense of Definition 4.38. See Theorem 2 of Rajput, B. S. & Cambanis, S. Gaussian processes and Gaussian measures. Ann. Math. Stat. 43, 1944–1952.

p204 The uniform convergence in (5.39) is a consequence of Mercer’s Theorem (Theorem 1.80). In particular, write $X(t) - X_J(t) = X(t) - \mu(t) - \sum_{j=1}^J \sqrt{\nu_j} \phi_j(t) \xi_j$. Then, $\mathbb{E}(X(t) - X_J(t))^2 = C(t, t) - \sum_{j=1}^J \nu_j |\phi_j(t)|^2 \geq 0$ from the KL expansion (as ξ_j has mean zero and unit variance). This converges to zero as $J \rightarrow \infty$ uniformly in t by Mercer’s Theorem (or directly by Lemma 1.78 as C and ϕ_j are continuous).

Chapter 6. **p235** In Algorithm 6.3, the normalisation by variance in Δv in the increments $\Delta \mathcal{W}_j$ is missing. To account for this, as $f(v_j) \Delta v_j = (\ell/2\pi) \times 2(\pi/\ell)/(J-1) = \frac{1}{J-1}$, the last line should be `Z=Z*sqrt(1/(J-1))`. See also `quad_sinc.m`.

¹<https://github.com/tonyshardlow/picspde>

p239 In Equation (6.34), the outer sum over k has not been carried over from the previous line. The equation should read

$$\tilde{Z}_R(t_n) = \sum_{k=0}^{N-1} e^{i2\pi kn/N} \sum_{m=0}^{M-1} e^{i(-R+m\Delta\nu)t_n} \sqrt{f(v_{kM+m})} \Delta\mathcal{W}_{kM+m}.$$

p279 The first two entries in the second column of the matrix V displayed above Example 7.41 should be $\tilde{u}_{2n_1+1}$ and $\tilde{u}_{2n_1+2}$, not $\tilde{u}_{2n_2+1}$ and $\tilde{u}_{2n_2+2}$.

Chapter 7. **p291** In Algorithm 7.10 (`turn_band_wm.m`), f should be an even function of s (missing modulus).

p305 In Lemma 7.67, the quantity $M_0(u)$ is infinite as stated. The proof of Theorem 7.68 can be fixed by using the Garsia–Rodemich–Rumsey inequality instead of Lemma 7.67.

Chapter 8.

Chapter 9.

Chapter 10. **p442–469** `meshgrid` is misused in Examples 10.12 and 10.40 and in Algorithms 10.5 and 10.10. See `exa_10.40.m`.

Appendix.

p487 The discrete Gronwall inequality (Lemma A.14) is incorrect. It should either be

Lemma 1. Consider $z_n \geq 0$ such that $z_n \leq a + bz_{n-1}$ for $n = 1, 2, \dots$ and $a, b \geq 0$. If $b = 1$, then $z_n \leq z_0 + na$. If $b \neq 1$, then

$$z_n \leq b^n z_0 + \frac{a}{1-b}(1-b^n).$$

or

Lemma 2. Consider $z_n \geq 0$ such that

$$z_n \leq a + b \sum_{k=0}^{n-1} z_k, \quad \text{for } n = 0, 1, 2, \dots \quad (*)$$

and constants $a, b \geq 0$. Then, $z_n \leq a(1+b)^n \leq a \exp(bn)$.

The first lemma is (Stuart and Humphries, 1997, Theorem 1.1.12).

To prove the second lemma, notice it is true for $n = 0$. Assume it is true for $z_0, \dots, z_{n-1}$. Then,

$$\begin{aligned} z_n &\leq a + b \sum_{k=0}^{n-1} z_k \quad \text{by } (*) \\ &\leq a + b \sum_{k=0}^{n-1} a(1+b)^k \quad \text{by induction assumption} \\ &= a + ab \frac{1 - (1+b)^n}{1 - (1+b)} \quad \text{by geometric summation formula} \\ &\leq a + ab \frac{1 - (1+b)^n}{-b} = a + ab((1+b)^n - 1) = a(1+b)^n. \end{aligned}$$

Therefore, $z_n \leq a(1+b)^n$ for all $n = 0, 1, 2, \dots$ by induction. Finally, $z_n \leq a \exp(bn)$ as $1+x \leq \exp(x)$ for $x \geq 0$. The second lemma can be used in the proof of Theorems 3.55 and 10.34.